

CustomerCluster

A customer segmentation and behavioral analytics platform built with Python, Streamlit, RFM analysis, and K-Means clustering.

Project Domain Customer analytics and business intelligence	Core Method RFM feature engineering + K-Means clustering
Input Data Online Retail transaction dataset	Output Segments, dashboards, and exportable tables
Use Case Retention planning, premium targeting, and segment review	Interface Interactive Streamlit dashboard with visual analytics

PREPARED BY

Abilash Amyreddy
AIML-A6/APRIL-9756
abilashreddycvr@gmail.com

Minor Project – Artificial Intelligence

ABSTRACT

CustomerCluster converts raw retail transaction data into actionable customer intelligence. The system computes Recency, Frequency, and Monetary metrics for each customer, standardizes them, and applies K-Means clustering to identify distinct behavioral groups. The resulting segments are presented through a Streamlit dashboard that includes KPI cards, scatter plots, statistical summaries, data exploration, and segment-specific recommendations.

It also highlights how the customer groups can be used for practical business action, including retention planning, premium targeting, and review of high-value segments. The dashboard makes the output readable for non-technical users while still keeping the RFM workflow, clustering logic, and exportable results easy to inspect.

1. Introduction

Customer segmentation is one of the most practical uses of machine learning in retail analytics. Instead of treating all customers the same, the system groups them by behavior so the business can understand who buys frequently, who spends more, and who needs re-engagement.

CustomerCluster is built around that idea. It transforms raw transaction history into a structured customer intelligence dashboard that supports analysis, comparison, and presentation. The application is designed to be readable for non-technical users while still reflecting the underlying analytics pipeline clearly.

2. Problem Statement

Retail transaction datasets are typically noisy and difficult to interpret directly. They may contain missing customer IDs, cancelled invoices, unusual spending spikes, and highly uneven purchasing patterns. A useful segmentation system must clean these records and still produce stable, explainable groups.

The objective of this project is to create a dashboard that converts transaction data into segment-level insight. The output should identify high-value customers, expose behavioral differences between clusters, and present the findings in a form that can support retention planning, premium targeting, and business review.

3. Objectives

The project aims to clean the dataset, engineer customer-level metrics, generate clusters using an unsupervised model, and present the output in a usable dashboard. It also supports filtering and CSV export so the analysis can be reused outside the app when needed.

4. Methodology

The workflow starts with preprocessing. Invalid or incomplete records are removed, customer-level aggregates are computed, and RFM features are extracted. Those values are standardized to prevent scale imbalance, and a pre-trained K-Means model assigns each customer to a segment. The dashboard then summarizes the clusters through charts, tables, and recommendation panels.

5. Technologies and Stack

Domain	Technologies Used
Frontend Interface	Streamlit, custom CSS styling
Backend and Logic	Python 3.x
Machine Learning	Scikit-learn, Pandas, NumPy
Visualization	Plotly, Matplotlib, Seaborn
Analytics	RFM analysis, K-Means clustering

6. Implementation Walkthrough — Overview and Analytics

The first image page shows the launch view and the main comparison view. These two together establish the visual language of the dashboard and lead the user from context to cluster analysis.

5.1 Dashboard Landing and KPI Overview

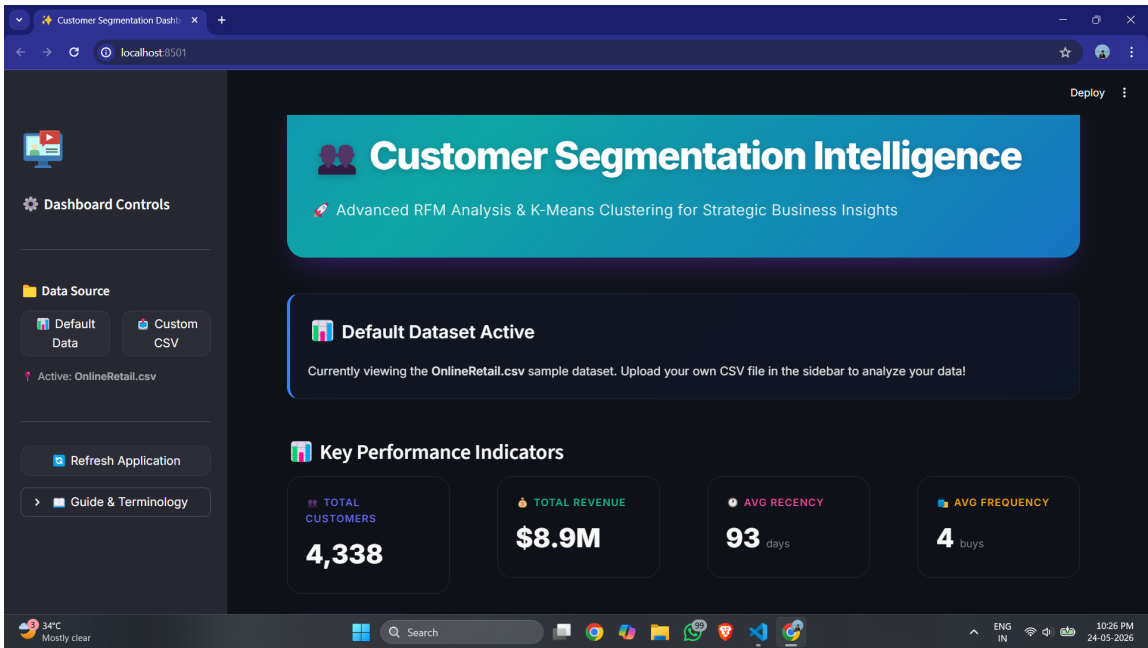


Figure 1: Main dashboard landing page and executive KPI overview.

The opening screen shows the project identity, active dataset, and headline KPIs. It provides the first business snapshot before the user enters detailed analytics.

5.2 Advanced Visual Analytics

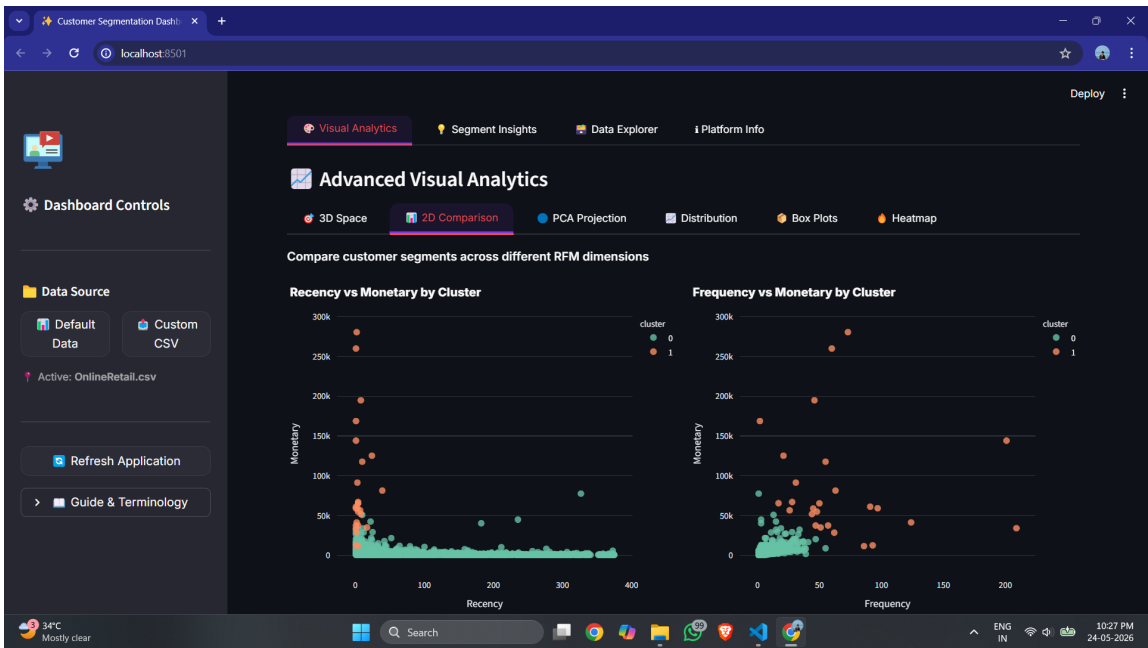


Figure 2: Cluster comparison and behavioral analytics visualizations.

This page compares customers across Recency, Frequency, and Monetary dimensions. It reveals how the main cluster differs from the high-value segment and makes the segmentation structure easy to see.

7. Implementation Walkthrough — Exploration and Summary

The second image page shows the explorer and the statistical summary. These views are used to validate the model output and inspect the customer table in detail.

5.3 Data Explorer and Export Module

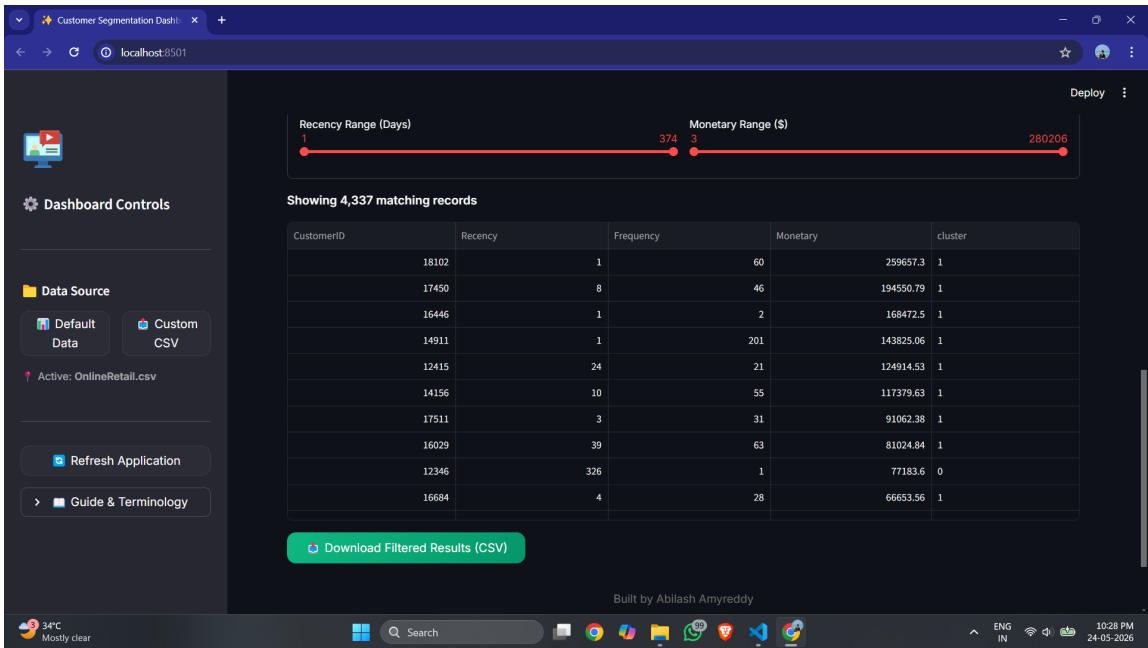


Figure 3: Customer-level data explorer with filtering and CSV export.

This module lets the user inspect the clustered customer table, narrow the data using filters, and export the result for reporting or external review.

5.4 Statistical Summary Dashboard

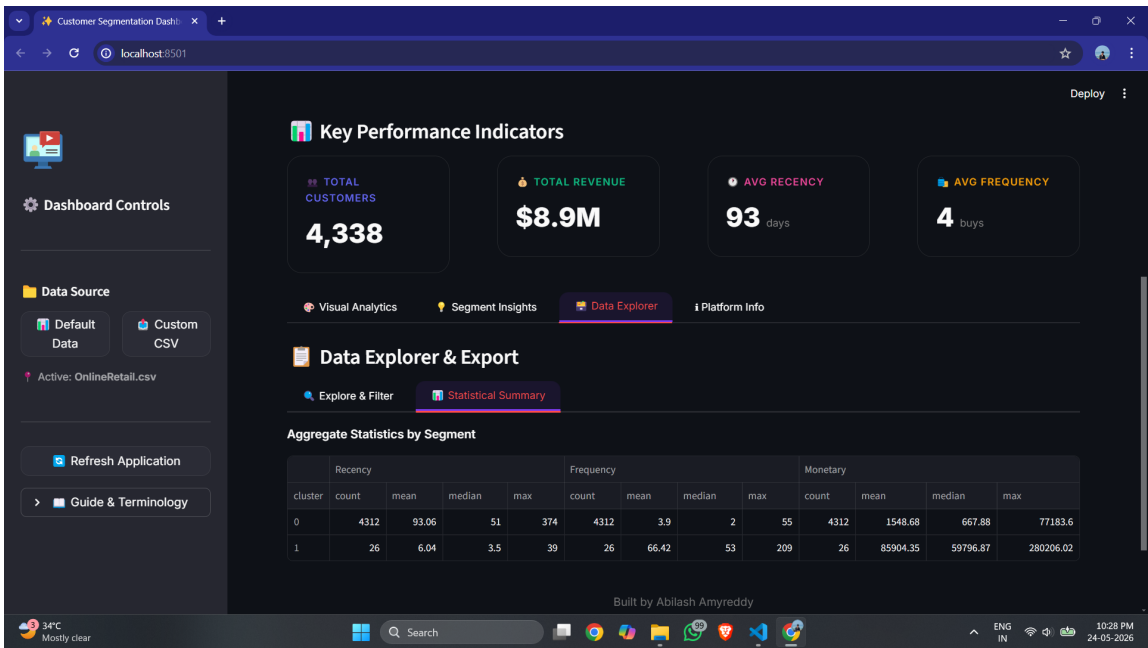


Figure 4: Statistical summaries and segment-level KPI analysis.

This view condenses cluster behavior into statistics such as mean, median, and maximum values. It is the numerical counterpart to the visual analytics pages.

8. Implementation Walkthrough — Architecture and Strategy

The final image page shows the technical architecture and the high-value segment page. This is where the report connects the model pipeline to strategic business decisions.

5.5 Platform Information and Technical Architecture

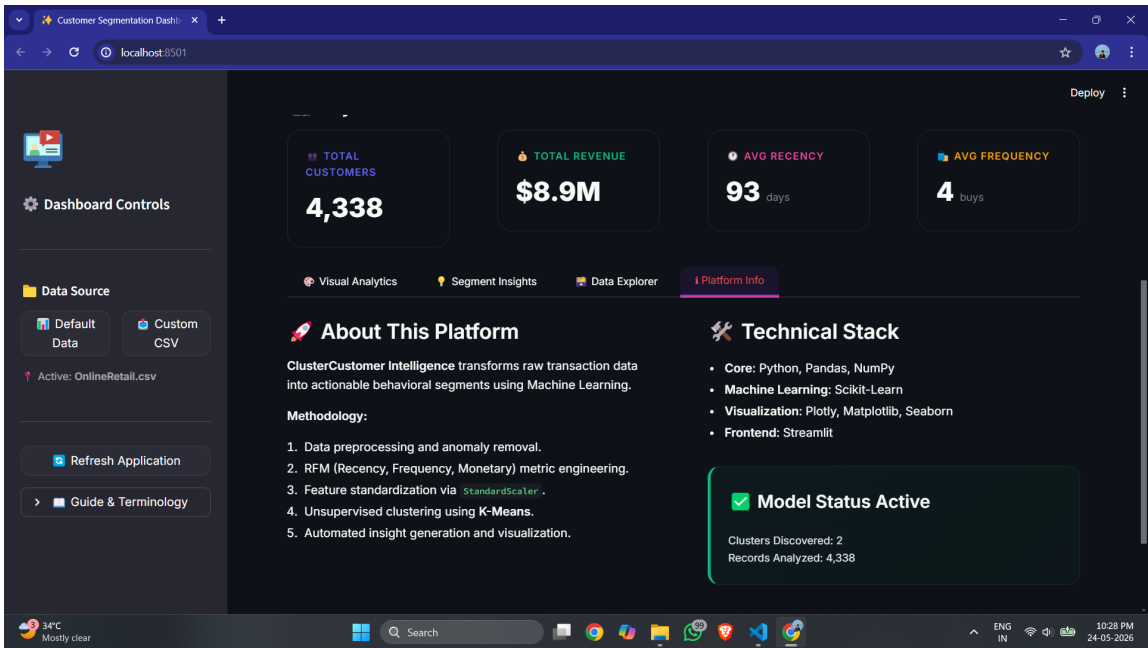


Figure 5: Platform methodology, workflow, and technical stack.

This section explains the preprocessing and clustering pipeline and documents the technologies used to build the application. It links the implementation back to the analytics process.

5.6 High-Value Customer Intelligence

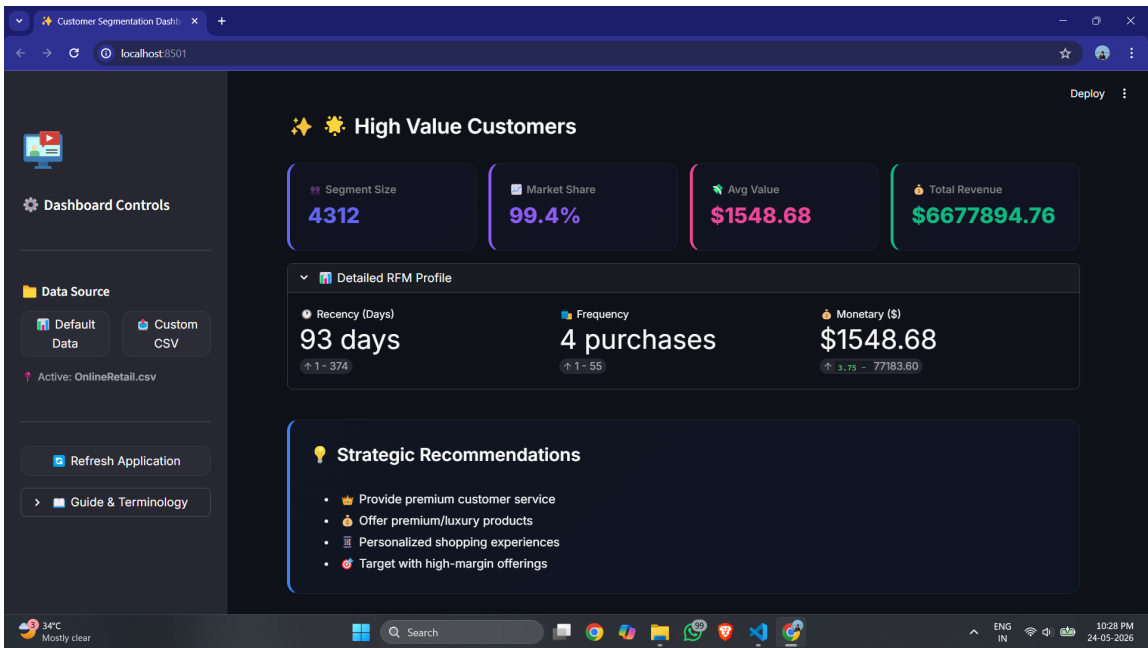


Figure 6: Premium customer segment insights and strategic recommendations.

This view focuses on the most valuable customer segment. It shows revenue contribution, average value, and suggested business actions such as premium service and retention targeting.

9. Results and Data Insights

The analysis produced a clear segmentation structure with a dominant low-value customer group and a smaller premium cluster. The dashboard summarizes 4,338 customers and approximately \$8.9 million in revenue, which makes the segmentation substantial enough for practical business use.

The strongest result is the identification of high-value customers who contribute disproportionately to revenue. That pattern is important because it shows that targeted service, premium offers, and retention effort can be concentrated where the business impact is highest.

The plots, table views, and segment summaries all reinforce the same message: customer behavior can be organized into measurable groups, and those groups can be translated into clear action.

10. Conclusion and Future Scope

Conclusion

CustomerCluster demonstrates how machine learning can be turned into a usable customer intelligence tool. By combining RFM analysis, clustering, and dashboard visualization, the project transforms raw transaction history into business insight that is easy to explain and easy to act on.

Future Scope

- integrate live customer databases for near real-time insight
- add advanced clustering methods such as DBSCAN or hierarchical clustering
- extend the app with churn prediction and customer lifetime value estimation
- automate retraining when new transaction data becomes available
- connect the output to marketing and CRM workflows

11. References

- Streamlit Documentation — docs.streamlit.io
- Scikit-learn Documentation — scikit-learn.org
- Pandas Documentation — pandas.pydata.org
- UCI Online Retail Dataset
- RFM Customer Segmentation Methodology